

Graduate Level - Short Answer Questions

1. Explain what SPA is and what DPA is. What are the differences between them?

SPA, stands for Simple Power Analysis, is the practice of inspecting a single power trace recorded while a device performs cryptographic operations, whereas DPA (Differential Power Analysis) is a statistical method of using many power traces recorded while the device processes different inputs.

SPA works better when the implementation has obvious yet secret dependent operations while the measurement being clean enough because only single or at most few power traces are recorded and analyzed. On the contrary, DPA works just fine even when individual traces are noisy because there are huge number of signals and traces that serves as the source. When found a specific pattern and noise being reduced, DPA provides very good comparative analysis that may reveal sensitive information.

2. Describe what tools and equipment are required to perform power analysis attacks.

As its name suggests, Power analysis attacks are a form of attack where attackers record, collect and analyze power traces from a system and use the analysis to reveal sensitive information like private keys of cryptographic operation, hardware's operational specs, and so on. Obviously, it requires variety of tools and equipment categorized from hardware to software.

High sample rated oscilloscope, EM probe for EM analysis, interface such as smartcard reader, signal generators/emulators are some of the hardware related tools and equipment used; whereas analysis tools such as ChipWhisperer, bundled software that comes with the hardware tool (oscilloscope), and other languages and libraries such as Numpy and SciPy for Python; are some that are software related.

In conjunction with both the above mentioned category, human behavior analysis and exploitation can be used as a complementary factor of successfully executing the overall attack.

3. Describe approaches of protecting a chip against power analysis attacks.

As we already know as of now, there is no single approach/tool/technique that can be used to prevent power analysis attack. Combinations of physical access restrictions, implementing masking tools correctly, EM shielding, sensor detectors, smooth power regulations irrespective of operation weight, etc. can be practically helpful to protect a chip against power analysis attacks.

Even though some approaches like EM shielding, implementing sensor detectors and smooth power regulations requires re-architecting the overall circuit design of a chip or its circuit board itself, a powerful design of such hardware along with simplest but effective methods such as physical restriction and general awareness can be used to improve protection against power analysis attacks.